

MODULE 4

Military Analytics Dashboard Documentation

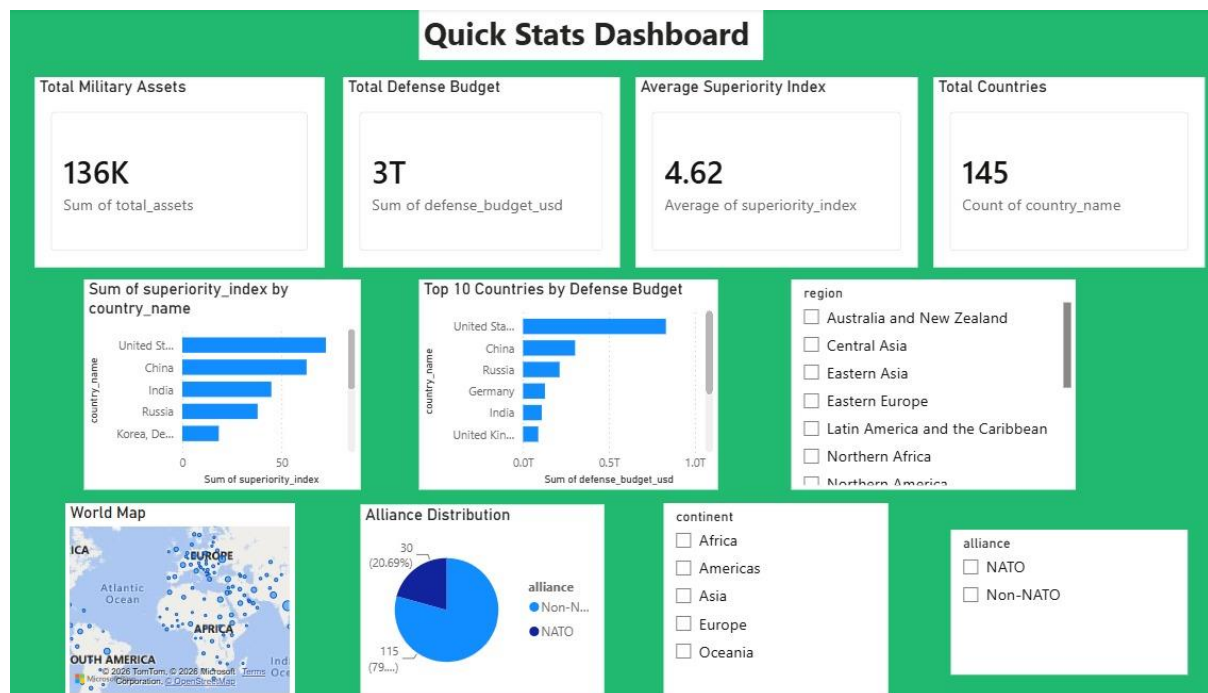
Project overview:

The Military Analytics Dashboard is a Power BI project developed to analyze and visualize global military data across multiple countries. The dashboard helps users understand defense budgets, military assets, personnel strength, geographical distribution, and comparative military power between nations.

Dashboard Overview:

- **Quick Stats Dashboard** : Provides a comprehensive overview of global military statistics through interactive KPI cards, charts, maps, and filters, allowing users to explore military data across countries, regions, continents, and alliances.
- **Nation Overview Dashboard** : Designed to present a detailed military profile for an individual country, including defense budget, population, manpower, military assets, rankings, and complete country-specific information through interactive visualizations.
- **Compare Powers Dashboard** : Intended to enable side-by-side comparison of two countries by evaluating key military metrics such as defense budget, manpower, aircraft, naval strength, and military superiority index.
- **Coalition Builder Dashboard** : Planned to simulate military alliances by allowing users to combine multiple countries into a coalition and compare their aggregated military capabilities against a selected reference country for strategic analysis.

Dashboard 1 : Quick Stats



Dashboard Components

KPI Cards

The dashboard begins with four summary cards that display the overall statistics of the dataset.

KPI	Description
Total Military Assets	Displays the total count of military assets across all countries.
Total Defense Budget	Shows the combined defense expenditure of all countries.
Average Superiority Index	Displays the average military superiority score calculated from the dataset.
Total Countries	Shows the total number of countries included in the analysis.

These KPIs update dynamically whenever filters are applied.

Superiority Index by Country

This horizontal bar chart displays countries ranked according to their military superiority index.

Purpose:

- Identify militarily stronger countries.
- Compare superiority scores.
- Highlight leading military powers.

Top 10 Countries by Defense Budget

This visualization ranks the ten countries with the largest defense budgets.

Purpose:

- Compare military expenditure.
- Identify highest defense investors.
- Analyze spending distribution.

World Map

The world map visualizes all countries geographically.

Purpose:

- Display global military distribution.
- Identify regional concentration.
- Improve geographical understanding of the dataset.

Alliance Distribution

A pie chart shows the proportion of NATO and Non-NATO countries.

Purpose:

- Understand alliance membership.
- Compare alliance distribution.
- Analyze geopolitical balance.

Interactive Filters

The dashboard contains several slicers that allow users to interact with the data.

Region Filter:

Allows users to filter countries by geographical region.

Examples:

- Eastern Asia
- Northern America
- Western Europe
- Southern Asia

Continent Filter:

Users can display data for a specific continent.

Options include:

- Africa
- Asia
- Europe
- Americas
- Oceania

Alliance Filter:

Users can compare:

- NATO countries
- Non-NATO countries

User Interaction

The dashboard supports interactive filtering.

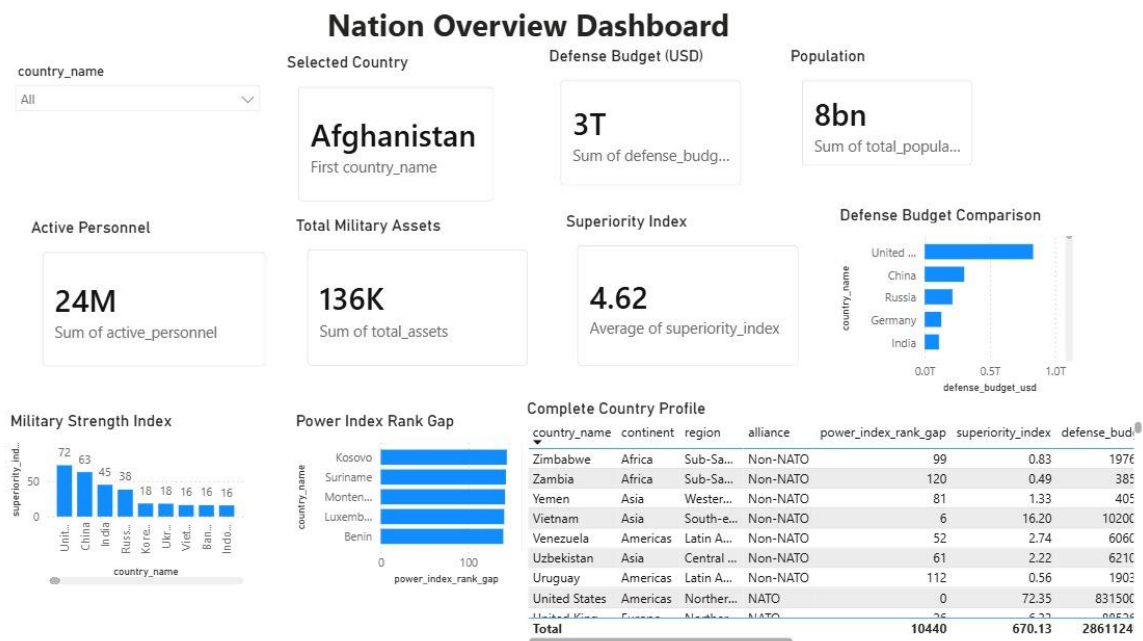
Selecting any filter automatically updates:

- KPI cards

- Charts
- World Map
- Alliance Distribution
- Defense Budget Rankings

This provides users with real-time analysis without refreshing the report.

Dashboard 2 : Nation Overview



Objective

The dashboard will allow users to select a country and instantly view its complete military profile.

Dashboard Components

Country Selection

A dropdown menu will allow users to select any country from the dataset.

Once selected, all visualizations will update automatically.

Country Summary

The dashboard will display:

- Country Name
- Defense Budget
- Population
- Active Personnel

- Total Military Assets
- Superiority Index

Defense Budget Comparison

A bar chart will compare the selected country's defense budget with leading military powers.

Purpose:

- Compare military spending.
- Identify global ranking.

Military Strength Index

A visualization showing:

- Military strength score
- Overall superiority
- Relative performance

Power Index Rank Gap

This chart will display the difference between the selected country's military ranking and other countries.

Purpose:

- Measure competitiveness.
- Understand ranking differences.

Complete Country Profile

A detailed table will display:

- Country
- Continent
- Region
- Alliance
- Power Index
- Superiority Index
- Defense Budget
- Additional military statistics

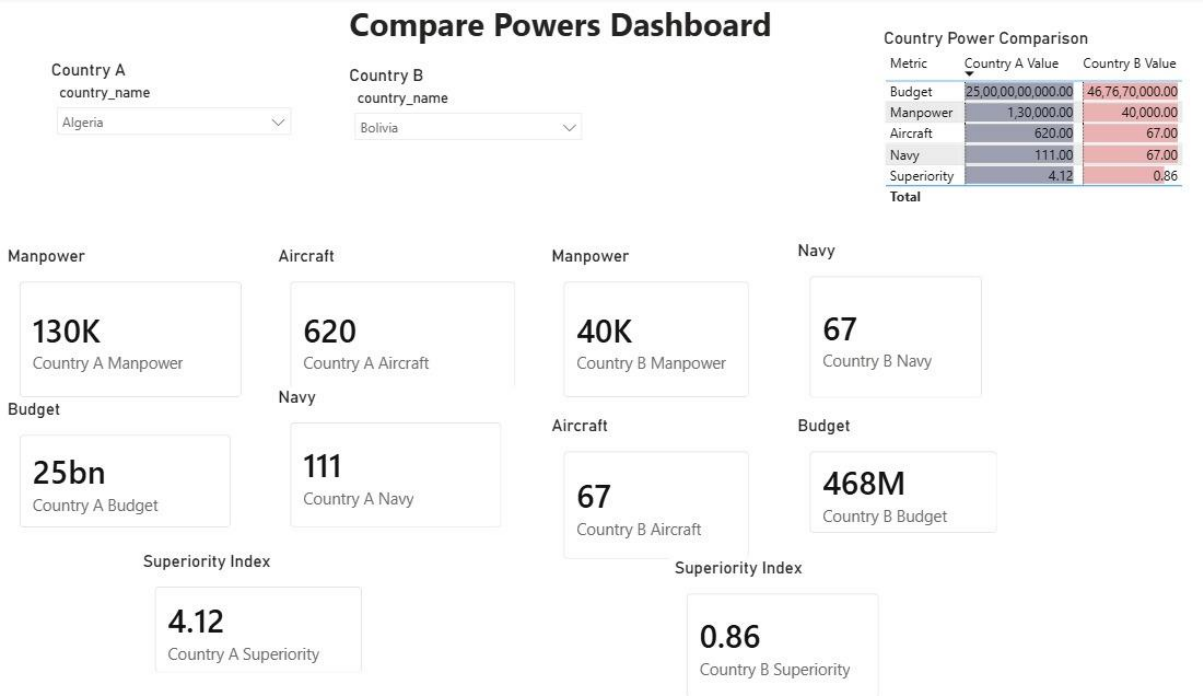
User Benefits

The Nation Overview Dashboard will enable users to:

- Analyze individual countries in detail.

- Study military strength and Compare rankings.
- Review complete military profiles.

Dashboard 3 : Compare Powers Dashboard



Objective

The objective is to simplify military comparison between any two selected countries.

Country Selection

Two dropdown lists will allow users to choose:

- Country A
- Country B

Individual KPI Cards

Each country will display:

- Defense Budget
- Manpower
- Aircraft
- Navy
- Superiority Index

Comparison Matrix

A comparison table will summarize major military metrics.

Metrics include:

- Budget
- Manpower
- Aircraft
- Navy
- Superiority Index

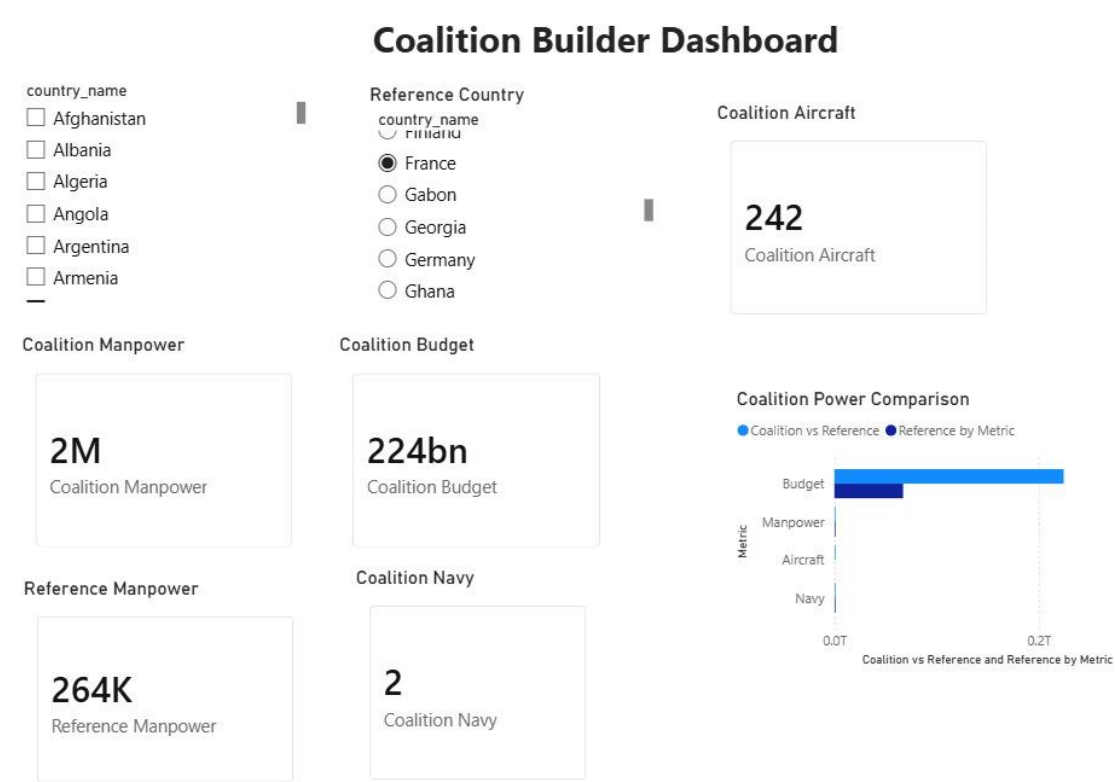
This table will help users quickly identify which country has stronger capabilities.

User Benefits

The dashboard will allow users to:

- Compare military resources.
- Evaluate strengths and weaknesses.
- Analyze defense spending.
- Support strategic decision-making.

Dashboard 4 : Coalition Builder



Objective

To provide a scenario-based analysis tool for evaluating the combined strength of military coalitions.

Coalition Selection

Users will select multiple countries to build a coalition.

The dashboard will aggregate military statistics for all selected countries.

Reference Country Selection

Users will select a single country to compare against the coalition.

Coalition KPI Cards

The dashboard will calculate combined values for:

- Coalition Manpower
- Coalition Budget
- Coalition Aircraft
- Coalition Navy

Reference Country KPIs

The dashboard will display:

- Reference Manpower
- Reference Budget
- Reference Aircraft
- Reference Navy

Coalition Power Comparison

A comparison chart will visualize:

- Coalition Strength
- Reference Country Strength

Comparison metrics include:

- Budget
- Manpower
- Aircraft
- Navy

User Benefits

The Coalition Builder Dashboard will allow users to:

- Simulate military alliances.

- Evaluate coalition strength.
- Compare multiple countries against a single nation.
- Support strategic military analysis and scenario planning.

Dashboard	Description
Quick Stats Dashboard	Provides an interactive overview of global military data through KPI cards, charts, maps, and filters. Users can analyze defense budgets, military assets, superiority index, and alliance distribution across countries.
Nation Overview Dashboard	Designed to display a detailed military profile of a selected country with key performance indicators and comparison charts. It includes visualizations for defense budget, manpower, military strength, rankings, and country-specific information.
Compare Powers Dashboard	Designed to compare the military capabilities of two countries side by side using key metrics. It includes comparison cards and a summary table for defense budget, manpower, aircraft, navy, and superiority index.
Coalition Builder Dashboard	Designed to analyze the combined military strength of multiple countries forming a coalition against a selected reference country. It includes aggregated coalition metrics and comparison visuals to support strategic analysis.